

Lab 15

Name : inshal nasir

Section : A

Roll no : 38

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Task : 1

```
variables.tf X locals.tf main.tf terraform.tfvars ...  
variables.tf  
1 variable "vpc_cidr_block" {}  
2 variable "subnet_cidr_block" {}  
3 variable "availability_zone" {}  
4 variable "env_prefix" {}  
5 variable "instance_type" {}  
6 variable "public_key_location" {}  
7 variable "private_key_location" {}
```

```
variables.tf locals.tf main.tf terraform.tfvars X ...  
terraform.tfvars  
1 vpc_cidr_block = "10.0.0.0/16"  
2 subnet_cidr_block = "10.0.10.0/24"  
3 availability_zone = "us-east-1a"  
4 env_prefix = "dev"  
5 instance_type = "t2.micro"  
6 public_key_location = "~/.ssh/id_rsa.pub"  
7 private_key_location = "~/.ssh/id_rsa"
```

```
variables.tf locals.tf X main.tf terraform.tfvars ...  
locals.tf  
1 data "http" "my_ip" {  
2 | url = "https://icanhazip.com"  
3 | }  
4  
5 locals {  
6 | my_ip = "${chomp(data.http.my_ip.response_body)}/32"  
7 | }
```

```
variables.tf  locals.tf  main.tf  terraform.tfvars

main.tf
1 resource "aws_vpc" "myapp-vpc" {
2   cidr_block = var.vpc_cidr_block
3   tags = {
4     Name = "${var.env_prefix}-vpc"
5   }
6 }
7
8 resource "aws_subnet" "myapp-subnet-1" {
9   vpc_id      = aws_vpc.myapp-vpc.id
10  cidr_block   = var.subnet_cidr_block
11  availability_zone = var.availability_zone
12  map_public_ip_on_launch = true # Taake servers ko public IP mile
13  tags = {
14    Name = "${var.env_prefix}-subnet-1"
15  }
16 }
17
18 resource "aws_internet_gateway" "myapp-igw" {
19   vpc_id = aws_vpc.myapp-vpc.id
20   tags = {
```

```
Generating public/private rsa key pair.
Created directory '/home/codespace/.ssh'.
Your identification has been saved in /home/codespace/.ssh/id_rsa
Your public key has been saved in /home/codespace/.ssh/id_rsa.pub
The key fingerprint is:
SHA256:BQaFtiPS71bv00AekKeD7l6SqtDxKbgfPZCkdNzYow codespace@codespaces-631a52
The key's randomart image is:
+---[RSA 4096]-----+
|      .++          |
|      o.=          |
|    . o E O..      |
|    . *. * B.+     |
|    oo*.OS         |
|    .O +* o        |
|    .+ +O.+ o      |
|    .O o .+.O..    |
| +O+  o. .O.      |
+---[SHA256]-----+
```

Initializing the backend...

Initializing provider plugins...

- Finding latest version of hashicorp/http...
- Finding latest version of hashicorp/aws...
- Installing hashicorp/http v3.5.0...
- Installed hashicorp/http v3.5.0 (signed by HashiCorp)
- Installing hashicorp/aws v6.28.0...
- Installed hashicorp/aws v6.28.0 (signed by HashiCorp)

Terraform has created a lock file `.terraform.lock.hcl` to record the provider selections it made above. Include this file in your version control repository so that Terraform can guarantee to make the same selections by default when you run "terraform init" in the future.

Terraform has been successfully initialized!

You may now begin working with Terraform. Try running "terraform plan" to see any changes that are required for your infrastructure. All Terraform commands should now work.

If you ever set or change modules or backend configuration for Terraform, rerun this command to reinitialize your working directory. If you forget, other commands will detect it and remind you to do so if necessary.

data.http.my_ip: Read complete after 0s [id=https://icanhazip.com]

Terraform used the selected providers to generate the following execution plan. Resource actions are indicated with the following symbols:

+ create

Terraform will perform the following actions:

```
# aws_internet_gateway.myapp-igw will be created
+ resource "aws_internet_gateway" "myapp-igw" {
  + arn      = (known after apply)
  + id       = (known after apply)
  + owner_id = (known after apply)
  + region   = "us-east-1"
  + tags     = {
    + "Name" = "dev-igw"
  }
  + tags_all = {
    + "Name" = "dev-igw"
  }
  + vpc_id   = (known after apply)
}
```

aws_route_table_association.a-rtb-subnet: Creation complete after 1s [id=rtbassoc-0d76a40660a03d42b]

Apply complete! Resources: 6 added, 0 changed, 0 destroyed.

Task : 2

 main.tf

```
72 resource "aws_key_pair" "ssh-key" {
73   key_name   = "server-key"
74   public_key = file(var.public_key_location)
75 }
76
77 data "aws_ami" "latest-amazon-linux-image" {
78   most_recent = true
79   owners      = ["amazon"]
80   filter {
81     name   = "name"
82     values = ["amzn2-ami-hvm-*-x86_64-gp2"]
83   }
84 }
85
86 resource "aws_instance" "frontend" {
87   ami                  = data.aws_ami.latest-amazon-linux-image.id
88   instance_type        = var.instance_type
89   subnet_id            = aws_subnet.myapp-subnet-1.id
90   vpc_security_group_ids = [aws_security_group.myapp-sg.id]
91   availability_zone     = var.availability_zone
92   associate_public_ip_address = true
93   key_name              = aws_key_pair.ssh-key.key_name
94
95   tags = {
```

 variables.tf U

 locals.tf U

 main.tf U

 outputs.tf U ×

 terraform.tfvars U

  ...

 outputs.tf

```
1  output "frontend_public_ip" {
2    | value = aws_instance.frontend.public_ip
3  }
4
5  output "backend_public_ips" {
6    | value = [for b in aws_instance.backend : b.public_ip]
7  }
8
9  output "backend_private_ips" {
10   | value = [for b in aws_instance.backend : b.private_ip]
11 }
```

```

XYjZttcLwi/1o3qQc3uLUR/lrOFj14RjflazNqojrZyq5/QzUy1SIEGHxZCD5PI+sCqRaGD/7OuEFg9ij1S4CwG/VG0djbNRT5Hs12CZ1ro7jY+R6ou
Md2faSYHI59uyvbB+vDlnA6Q1CaPpdMM1GX3+IjJIMfVrCw9e6vNYLktIvQ== codespace@codespaces-631a52"
+ region      = "us-east-1"
+ tags_all    = (known after apply)
}

```

Plan: 5 to add, 0 to change, 0 to destroy.

Changes to Outputs:

```

+ backend_private_ips = [
+   (known after apply),
+   (known after apply),
+   (known after apply),
+ ]
+ backend_public_ips = [
+   (known after apply),
+   (known after apply),
+   (known after apply),
+ ]
+ frontend_public_ip = (known after apply)

```

Note: You didn't use the -out option to save this plan, so Terraform can't guarantee to take exactly these actions if you run "terraform apply" now.

Apply complete! Resources: 5 added, 0 changed, 0 destroyed.

Outputs:

```

backend_private_ips = [
  "10.0.10.139",
  "10.0.10.168",
  "10.0.10.7",
]
backend_public_ips = [
  "44.222.93.92",
  "98.92.54.207",
  "34.205.166.198",
]
frontend_public_ip = "44.195.90.205"

```

Task : 3

variables.tf U
locals.tf U
main.tf U
outputs.tf U
ansible.cfg U ×
terraform.tfvars U

```

ansible > ansible.cfg
1 [defaults]
2 host_key_checking = False
3 interpreter_python = /usr/bin/python3
4 inventory = ./inventory/hosts

```

variables.tf U
locals.tf U
main.tf U
outputs.tf U
ansible.cfg U
hosts U ×
terraform.tfvars U

```

ansible > inventory > hosts
1 [frontend]
2 44.195.90.205
3
4 [backends]
5 44.222.93.92
6 98.92.54.207
7 34.205.166.198
8
9 [all:vars]
10 ansible_user=ec2-user
11 ansible_ssh_private_key_file=~/.ssh/id_rsa

```

```

98.92.54.207 | SUCCESS => {
  "changed": false,
  "ping": "pong"
}
44.222.93.92 | SUCCESS => {
  "changed": false,
  "ping": "pong"
}
44.195.90.205 | SUCCESS => {
  "changed": false,
  "ping": "pong"
}
34.205.166.198 | SUCCESS => {
  "changed": false,
  "ping": "pong"
}
}

```

Task : 4

ansible > roles > backend > tasks > ! main.yml

```

1  ---
2  - name: Install Apache (httpd)
3    yum:
4      name: httpd
5      state: present
6      update_cache: yes
7
8  - name: Enable and start httpd
9    service:
10     name: httpd
11     state: started
12     enabled: true
13
14 - name: Deploy backend index page from template
15   template:
16     src: backend_index.html.j2
17     dest: /var/www/html/index.html
18     owner: apache
19     group: apache
20     mode: '0644'
21   notify: restart httpd

```

f U main.tf U outputs.tf U ansible.cfg U hosts U ! main.yml U backend_index.html.j2 U ×

ansible > roles > backend > templates > backend_index.html.j2

```

1  <!DOCTYPE html>
2  <html>
3  <head>
4    <title>Backend {{ inventory_hostname }}</title>
5  </head>
6  <body>
7    <h1>Backend server: {{ inventory_hostname }}</h1>
8    <p>Public IP: {{ inventory_hostname }}</p>
9    <p>Private IP: {{ ansible_default_ipv4.address | default('unknown') }}</p>
10   <p>Welcome to the backend cluster!</p>
11 </body>
12 </html>

```

ts.tf U ansible.cfg U hosts U ! main.yml .../tasks U backend_index.html.j2 U ! main.yml .../handlers U ×

ansible > roles > backend > handlers > ! main.yml

```

1  ---
2  - name: restart httpd
3    service:
4      name: httpd
5      state: restarted

```

```
PLAY [Deploy Apache to Backend Servers] *****

TASK [Gathering Facts] *****
ok: [44.222.93.92]
ok: [34.205.166.198]
ok: [98.92.54.207]

TASK [backend : Install Apache (httpd)] *****
changed: [44.222.93.92]
changed: [98.92.54.207]
changed: [34.205.166.198]

TASK [backend : Enable and start httpd] *****
changed: [44.222.93.92]
changed: [34.205.166.198]
changed: [98.92.54.207]

TASK [backend : Deploy backend index page from template] *****
changed: [44.222.93.92]
changed: [34.205.166.198]
changed: [98.92.54.207]

RUNNING HANDLER [backend : restart httpd] *****
changed: [44.222.93.92]
changed: [34.205.166.198]
changed: [98.92.54.207]

PLAY RECAP *****
34.205.166.198      : ok=5    changed=4    unreachable=0    failed=0    skipped=0    rescued=0    ignored=0
44.222.93.92       : ok=5    changed=4    unreachable=0    failed=0    skipped=0    rescued=0    ignored=0
98.92.54.207       : ok=5    changed=4    unreachable=0    failed=0    skipped=0    rescued=0    ignored=0
```

Backend server: 98.92.54.207

Public IP: 98.92.54.207

Private IP: 10.0.10.168

Welcome to the backend cluster!

Backend server: 34.205.166.198

Public IP: 34.205.166.198

Private IP: 10.0.10.7

Welcome to the backend cluster!

Backend server: 44.222.93.92

Public IP: 44.222.93.92

Private IP: 10.0.10.139

Welcome to the backend cluster!

Task : 5

ansible > roles > frontend > templates > nginx_frontend.conf.j2

```
1 user nginx;
2 worker_processes auto;
3 error_log /var/log/nginx/error.log notice;
4 pid /run/nginx.pid;
5
6 events {
7     worker_connections 1024;
8 }
9
10 http {
11     include /etc/nginx/mime.types;
12     default_type application/octet-stream;
13
14     upstream backend_servers {
15         # Pehle do servers normal challenge
16         server {{ backend1_private_ip }}:80;
```

```
1 ---
2 - name: Install nginx
3   yum:
4     name: nginx
5     state: present
6     update_cache: yes
7
8 - name: Enable and start nginx
9   service:
10    name: nginx
11    state: started
12    enabled: true
13
14 - name: Deploy nginx frontend config
15   template:
16     src: nginx_frontend.conf.j2
17     dest: /etc/nginx/nginx.conf
18   notify: Restart nginx
```

ansible > roles > frontend > handlers > ! main.yml

```
1 ---
2 - name: Restart nginx
3   service:
4     name: nginx
5     state: restarted
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```



```
PLAY [Setup Load Balancer] *****

TASK [Gathering Facts] *****
ok: [44.195.90.205]

TASK [frontend : Enable Nginx from Amazon Linux Extras] *****
changed: [44.195.90.205]

TASK [frontend : Install nginx] *****
changed: [44.195.90.205]

TASK [frontend : Enable and start nginx] *****
changed: [44.195.90.205]

TASK [frontend : Deploy nginx frontend config] *****
changed: [44.195.90.205]

RUNNING HANDLER [frontend : Restart nginx] *****
changed: [44.195.90.205]

PLAY RECAP *****
44.195.90.205      : ok=6    changed=5    unreachable=0    failed=0    skipped=0    rescued=0    ignored=0
```

Backend server: 44.222.93.92

Public IP: 44.222.93.92

Private IP: 10.0.10.139

Welcome to the backend cluster!

Task : 6

```
ansible > roles > trontena > templates > nginx_trontena.conf.j2
1  user nginx;
2  worker_processes auto;
3  error_log /var/log/nginx/error.log notice;
4  pid /run/nginx.pid;
5
6  events {
7      worker_connections 1024;
8  }
9
10 http {
11     include /etc/nginx/mime.types;
12     default_type application/octet-stream;
13
14     upstream backend_servers {
15         # Dynamically picking IPs from inventory
16         server {{ backend1_private_ip }}:80;
17         server {{ backend2_private_ip }}:80;
18         server {{ backup_backend_private_ip }}:80 backup;
19     }
20
21     server {
22         listen 80;
23         server_name _;
```



```
ansible > ! site.yml
```

```
1  ---
2  - name: Configure backend HTTPD servers
3    hosts: backends
4    become: true
5    roles:
6      - backend
7
8  - name: Configure frontend Nginx load balancer
9    hosts: frontend
10   become: true
11   vars:
12     # Requirement 6: Dynamic gathering using hostvars
13     backend1_private_ip: "{{ hostvars[groups['backends'][0]].ansible_default_ipv4.address }}"
14     backend2_private_ip: "{{ hostvars[groups['backends'][1]].ansible_default_ipv4.address }}"
15     backup_backend_private_ip: "{{ hostvars[groups['backends'][2]].ansible_default_ipv4.address }}"
16   roles:
17     - frontend
```

```
changed: [44.195.90.205]
```

```
TASK [frontend : Install nginx] *****
ok: [44.195.90.205]
```

```
TASK [frontend : Enable and start nginx] *****
ok: [44.195.90.205]
```

```
TASK [frontend : Deploy nginx frontend config] *****
changed: [44.195.90.205]
```

```
RUNNING HANDLER [frontend : Restart nginx] *****
changed: [44.195.90.205]
```

```
PLAY RECAP *****
34.205.166.198      : ok=4    changed=0    unreachable=0    failed=0    skipped=0    rescued=0    ignored=0
44.195.90.205      : ok=6    changed=3    unreachable=0    failed=0    skipped=0    rescued=0    ignored=0
44.222.93.92       : ok=4    changed=0    unreachable=0    failed=0    skipped=0    rescued=0    ignored=0
98.92.54.207       : ok=4    changed=0    unreachable=0    failed=0    skipped=0    rescued=0    ignored=0
```

Backend server: 44.222.93.92

Public IP: 44.222.93.92

Private IP: 10.0.10.139

Welcome to the backend cluster!

Backend server: 98.92.54.207

Public IP: 98.92.54.207

Private IP: 10.0.10.168

Welcome to the backend cluster!

Task : 7

.terraform >  hosts.tpl

```
1 [frontend]
2 ${frontend_ip}
3
4 [backends]
5 %{ for ip in backend_ips ~}
6 ${ip}
7 %{ endfor ~}
8
9 [all:vars]
10 ansible_user=ec2-user
11 ansible_ssh_private_key_file=~/.ssh/id_rsa
12 ansible_python_interpreter=/usr/bin/python3
```

 main.tf

```
114 resource "local_file" "ansible_inventory" {
115   content = templatefile("hosts.tpl", {
116     frontend_ip = aws_instance.frontend.public_ip
117     backend_ips = [for b in aws_instance.backend : b.public_ip]
118   })
119   filename = "ansible/inventory/hosts"
120 }
121
122 resource "null_resource" "ansible_config" {
123   triggers = {
124     frontend_ip = aws_instance.frontend.public_ip
125     backend_ips = join(",", [for b in aws_instance.backend : b.public_ip])
126   }
127
128   depends_on = [
129     aws_instance.frontend,
130     aws_instance.backend,
131     local_file.ansible_inventory
132   ]
133
134   provisioner "local-exec" {
135     command = <<-EOT
136     sleep 30;
137     cd ansible
138     ansible-playbook site.yml
139     EOT
140   }
141 }
```

```

ansible > roles > frontend > tasks > ! main.yml
4  - name: Enable nginx from Amazon Linux ELAS
5  - name: Install nginx
6    yum:
7      name: nginx
8      state: present
9      update_cache: yes
10
11 - name: Enable and start nginx
12   service:
13     name: nginx
14     state: started
15     enabled: true
16
17 - name: Deploy nginx frontend config
18   template:
19     src: nginx_frontend.conf.j2
20     dest: /etc/nginx/nginx.conf
21   notify: Restart nginx

```

Apply complete! Resources: 2 added, 0 changed, 0 destroyed.

Outputs:

```

backend_private_ips = [
  "10.0.10.139",
  "10.0.10.168",
  "10.0.10.7",
]
backend_public_ips = [
  "44.222.93.92",
  "98.92.54.207",
  "34.205.166.198",
]
frontend_public_ip = "44.195.90.205"

```

Backend server: 44.222.93.92

Public IP: 44.222.93.92

Private IP: 10.0.10.139

Welcome to the backend cluster!

Backend server: 98.92.54.207

Public IP: 98.92.54.207

Private IP: 10.0.10.168

Welcome to the backend cluster!

```
create mode 100644 ansible/site.yml
create mode 100644 hosts.tpl
create mode 100644 locals.tf
create mode 100644 main.tf
create mode 100644 outputs.tf
create mode 100644 variables.tf
```

```
Enumerating objects: 33, done.
Counting objects: 100% (33/33), done.
Delta compression using up to 2 threads
Compressing objects: 100% (25/25), done.
Writing objects: 100% (31/31), 4.96 KiB | 4.96 MiB/s, done.
Total 31 (delta 0), reused 28 (delta 0), pack-reused 0 (from 0)
To https://github.com/23-22411-016-droid/LabProject_FrontendBackend
9a551c6..0a7adac main -> main
```